

RC Cantilever Column: Lattice vs. Force-Based Beam–Column

Model, calibration, and static / dynamic analysis

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1 Overview

The study compares two backend idealizations of the *same* physical column (Section 2):

- **Reference model** — a force-based fiber `forceBeamColumn` cantilever (Section 3.1).
- **Lattice model** — a 2D assembly of axial truss struts generated on a structured `gms` node grid, with longitudinal and transverse reinforcement represented as steel struts on shared nodes (Section 3.2).

Four analysis cases are reported, each run for both models:

Case	Loading	Constitutive setting
Pushover (linear)	monotonic lateral displacement	Elastic
Pushover (nonlinear)	monotonic lateral displacement	Concrete02 + Steel02
Dynamic (linear)	base acceleration (sine, El Centro)	Elastic
Dynamic (nonlinear)	base acceleration (sine, El Centro)	Concrete02 + Steel02

The lattice strut area is calibrated so that the lattice and the fiber column share the same initial lateral stiffness K_0 (Section 4); this same calibration fixes the fundamental period used by the dynamic cases. All quantities use a consistent kip, inch, second unit system.

2 Physical specimen

The specimen is a vertical RC cantilever, fixed at the base and free at the top, carrying a constant axial compression at the top and loaded laterally in the plane of its larger section dimension. The specimen is identical for every analysis case.

2.1 Geometry and loading

Table 1: Specimen geometry and applied loads.

Quantity	Symbol	Value
Cantilever height	H	144 in
Section depth (in-plane, x)	W	24 in
Section width (out-of-plane)	t	15 in
Clear cover	–	1.5 in
Axial load (constant, top)	P	180 kip (compression)
Reference lateral load	H_{ref}	10 kip (shape only)
Base support	–	fully fixed (u_x, u_y)

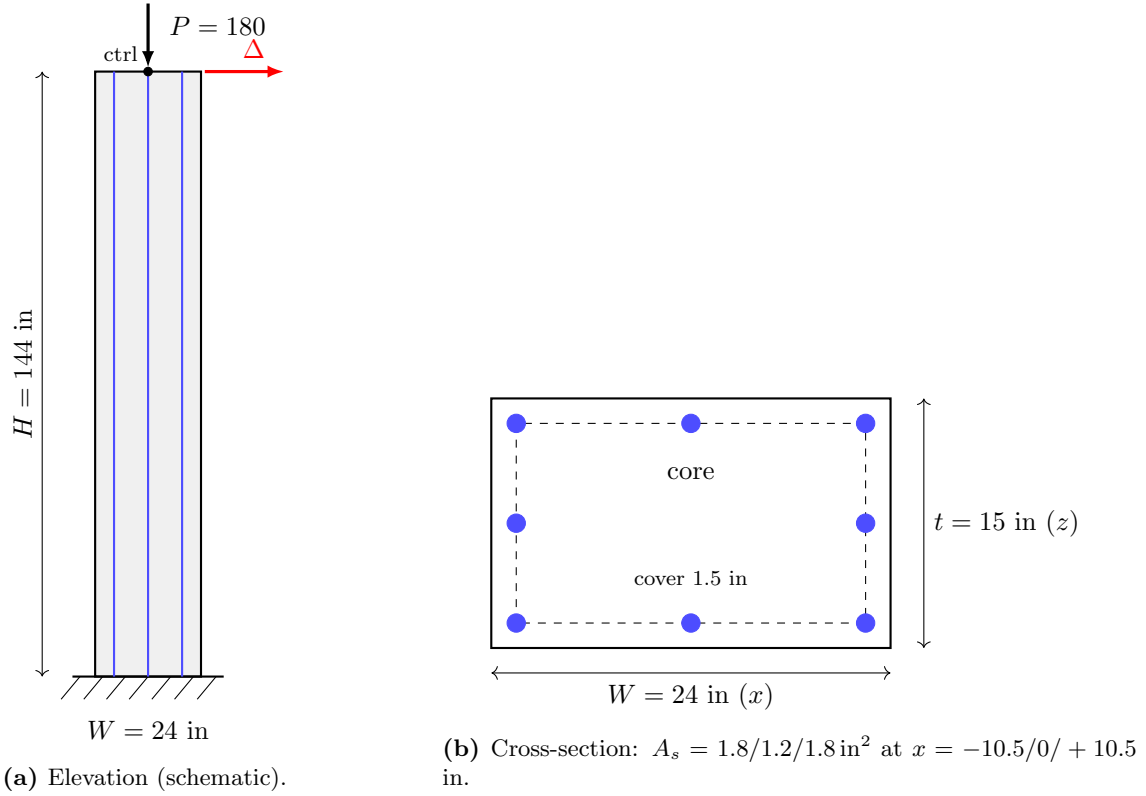


Figure 1: Physical specimen. Lateral load and displacement Δ act along x (bending about the z axis). Drawings are schematic; dimensions are labelled with their true values.

2.2 Materials

Concrete is defined by two physical grades — a *confined core* and an *unconfined cover* — and the reinforcement by a single steel grade. Values are stored as positive magnitudes; the OpenSees sign convention (compression negative) is applied in the material mapping.

Table 2: Concrete grades (physical input). ρ is mass density.

Grade	E (ksi)	ν	ρ (kip s ² /in ⁴)	f'_c (ksi)	ε_{c0}	f_{cu} (ksi)	ε_U	f_t (ksi)
Confined core	4030	0.20	2.25×10^{-7}	6.0	0.004	4.0	0.030	0.6
Unconfined cover	4030	0.20	2.25×10^{-7}	5.0	0.002	0.0	0.006	0.5

Table 3: Reinforcing-steel grade (Steel02 / Giuffré–Menegotto–Pinto).

Grade	f_y (ksi)	E_0 (ksi)	b	R_0	c_{R1}	c_{R2}
Reinforcement	60.0	30000	0.01	18.0	0.925	0.15

2.3 Reinforcement layout

Table 4: Reinforcement. Longitudinal bars lump the benchmark 3+2+3 bars at 0.6 in^2 each into one area per layer; transverse ties are placed one per mesh row.

Type	Position	Area (in^2)	Extent
Longitudinal layer 1	$x = -10.5\text{ in}$	1.8	full height $0 \leq y \leq 144$
Longitudinal layer 2	$x = 0\text{ in}$	1.2	full height $0 \leq y \leq 144$
Longitudinal layer 3	$x = +10.5\text{ in}$	1.8	full height $0 \leq y \leq 144$
Transverse ties	each mesh row y_i	0.3 per tie	$x = -10.5 \rightarrow +10.5$

Total longitudinal steel area is 4.8 in^2 , identical in both models.

3 Numerical models

3.1 Force-based fiber beam-column (reference)

The reference is a single force-based `forceBeamColumn` element with a fiber section, in a 2D model with three DOFs per node (u_x, u_y, θ_z).

Table 5: Fiber beam-column model definition.

Item	Specification
Model dimension	<code>ndm=2, ndf=3</code>
Nodes	base (fixed), top (free, control)
Element	<code>forceBeamColumn</code>
Geometric transform	<code>PDelta</code>
Beam integration	<code>Lobatto</code> , 5 integration points
Fiber section	$24\text{ (}y\text{)} \times 15\text{ (}z\text{)}$, cover 1.5 in
core patch	10×1 fibers (material: core)
cover patches	1-fiber strips (material: cover)
steel layers	3/2/3 bars at $y = -10.5/0/+10.5$, $A_s = 0.6\text{ in}^2$
Axial load	$P = 180\text{ kip}$ at the top node (held constant)

The fiber materials are *matched* to the lattice grades: nonlinear cases use `Concrete02` (core/cover) and `Steel02`; linear cases use `Elastic` ($E = 4030\text{ ksi}$ concrete, $E_0 = 30000\text{ ksi}$ steel). For the nonlinear dynamic case the column is subdivided into 96 equal force-based elements (nodes at the lattice row heights) so the self-mass distributes by tributary length identically to the lattice. The fiber-section `Concrete02` laws use the grade values directly (no length regularization; there is no fiber analog):

Table 6: Fiber-section **Concrete02** arguments ($f_{pc}, \varepsilon_{c0}, f_{pcu}, \varepsilon_U, \lambda, f_t, E_{ts}$), compression negative.

Fiber material	Arguments
Core (nonlinear)	(−6.0, −0.004, −4.0, −0.030, 0.1, 0.6, 403.0)
Cover (nonlinear)	(−5.0, −0.002, −0.0, −0.006, 0.1, 0.5, 403.0)
Steel (Steel02)	(60.0, 30000, 0.01, 18.0, 0.925, 0.15)

3.2 RC lattice model

The lattice is a 2D assembly of axial struts (two translational DOFs per node) built on a structured **gms**h grid; struts connect node pairs within a peridynamic-style horizon, and reinforcement is added as steel struts on the shared lattice nodes (perfectly bonded).

Table 7: RC lattice model definition.

Item	Specification
Model dimension	ndm=2, ndf=2
Mesh generator	gms h structured transfinite grid
Mesh size	1.5 in (16×96 cells)
Nodes	1649 (17×97 grid)
Strut element	corotTruss (corotational, geometric nonlinearity)
Strut connectivity	horizon rule, $r = \text{horizon} \times \text{mesh} = 1.5 \times 1.5 = 2.25$ in
orthogonal struts	length 1.5 in (vertical + horizontal)
diagonal struts	length $1.5\sqrt{2} = 2.1213$ in
Concrete material	per-zone length-regularized Concrete02 (4 distinct)
Rebar material	Steel02 (1 grade)
Base support	17 base-row nodes fixed (u_x, u_y)
Control node	top-center node

Table 8: Lattice element inventory (7902 struts total).

Concrete struts	count	Reinforcement struts	count
vertical (orthogonal)	1632	longitudinal (3 layers)	288
horizontal (orthogonal)	1552	transverse ties (97 rows)	1358
diagonal	3072		
concrete total	6256	rebar total	1646

Length-regularized concrete (crack-band). Both softening branches of the strut concrete law are regularized by the strut length L , so that the dissipated energy is mesh-objective:

$$E_{ts} = \frac{f_t^2 L}{2G_f} \quad (\leq 0.5E), \quad \varepsilon_U = \varepsilon_{c0} + \frac{2G_{fc}}{(f_c + f_{cu})L} \quad (\geq \varepsilon_U^{\text{grade}}),$$

with a residual compression plateau $f_{cu} \geq 0.2f_c$. The fracture energies are $G_f = 4.0 \times 10^{-3}$ kip/in (tension) and $G_{fc} = 1.5$ kip/in (compression). This yields four concrete materials (two zones $\times$ two strut lengths):

Table 9: Lattice Concrete02 arguments ($f_{pc}, \varepsilon_{c0}, f_{pcu}, \varepsilon_U, \lambda, f_t, E_{ts}$), compression negative.

Zone / strut length	Arguments
Cover, $L = 1.5$	(-5.0, -0.002, -1.0, -0.33533, 0.1, 0.5, 46.875)
Cover, $L = 2.1213$	(-5.0, -0.002, -1.0, -0.23770, 0.1, 0.5, 66.291)
Core, $L = 1.5$	(-6.0, -0.004, -4.0, -0.20400, 0.1, 0.6, 67.500)
Core, $L = 2.1213$	(-6.0, -0.004, -4.0, -0.14542, 0.1, 0.6, 95.459)
Rebar (Steel02)	(60.0, 30000, 0.01, 18.0, 0.925, 0.15)

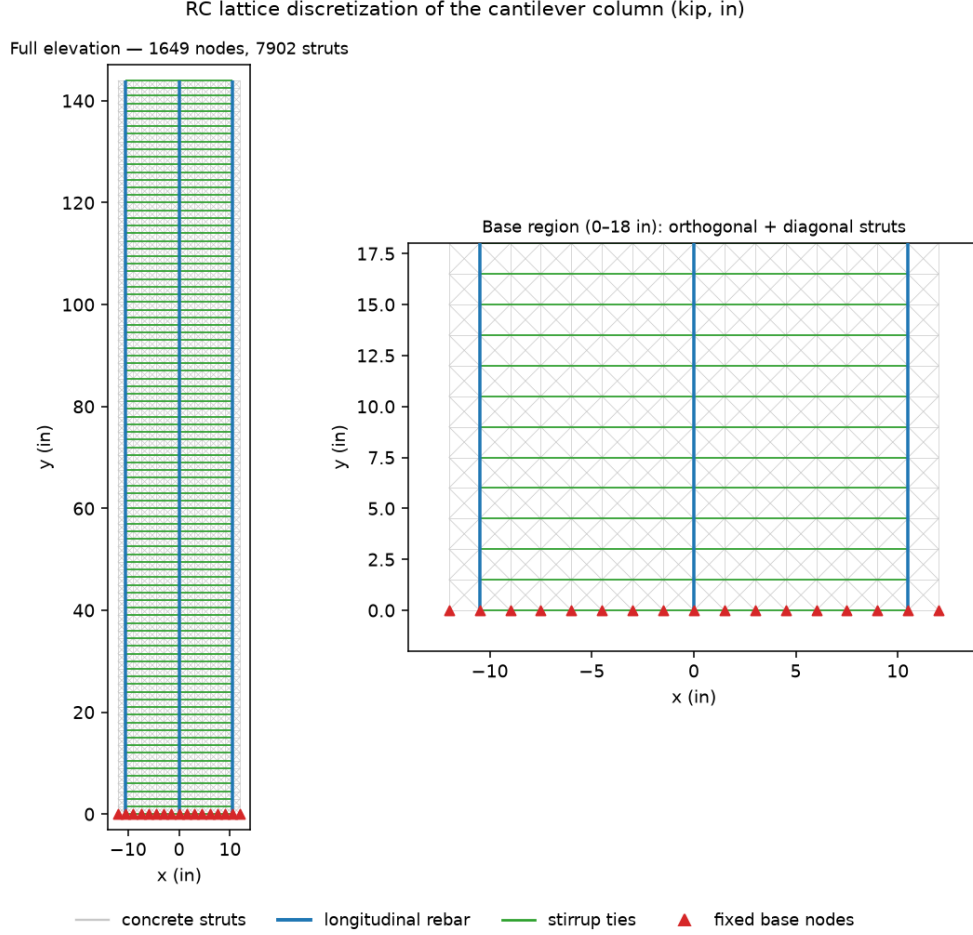


Figure 2: RC lattice discretization: concrete horizon struts (grey), longitudinal rebar struts (blue), transverse stirrup ties (green), and fixed base nodes (red). Left: full elevation; right: base region detail showing the orthogonal and diagonal strut pattern.

Mass. Nodal mass is lumped by tributary volume ($\rho t A_{\text{trib}}/4$ per quad corner) from the same grid used to draw the struts. The total lattice self-mass is $\sum m = 1.1664 \times 10^{-2} \text{ kip s}^2/\text{in}$ ($= \rho V$, $V = 24 \times 144 \times 15$). For the dynamic cases a top lumped mass $P/g = 0.4658 \text{ kip s}^2/\text{in}$ (the axial load as tributary seismic mass, $g = 386.4 \text{ in/s}^2$) is added at the top nodes.

4 Stiffness calibration

The single calibration parameter is the concrete strut cross-sectional area A (uniform over all concrete struts), chosen so the lattice's initial lateral stiffness K_0 equals the fiber column's. The

reinforcement areas are fixed by the specimen and are not calibrated.

- **Nonlinear cases:** a plain elastic lattice (reinforcement omitted) has $K_0 \propto A$, so a single solve gives A from the fiber-column K_0 .
- **Linear cases:** the full RC elastic lattice (concrete + rebar) has a stiffness that is monotone but not proportional in A (rebar adds a parallel stiffness), so A is found by a secant root-find on $K_0(A) - K_0^{\text{ref}}$.

Table 10: Calibration results. K_0^{ref} is the fiber-column initial lateral stiffness (first finite pushover step). The nonlinear K_0^{ref} is the **Concrete02** initial tangent; the linear value is the elastic stiffness.

Configuration	K_0^{ref} (kip/in)	Calibrated A (in ²)	Lattice K_0 (kip/in)	Error
Linear	68.69	11.736	68.69	−0.00%
Nonlinear	61.53	12.453	matched by construction	—

5 Analysis procedures and parameters

5.1 Pushover

A constant-gravity stage (axial P) is applied first and held; the lateral displacement of the control node is then increased monotonically under displacement control while the base shear (sum of horizontal base reactions) is recorded.

Table 11: Pushover analysis parameters (both models).

Item	Setting
Constraint handler	Transformation
DOF numberer	RCM
System of equations	BandGeneral
Gravity stage	LoadControl , 10 steps; Newton ; NormDispIncr 10^{-8}
Hold gravity	loadConst (time reset to 0)
Lateral stage	DisplacementControl on control DOF, $dU = 0.05$ in
Target tip displacement	10.0 in
Convergence test	NormDispIncr , tol 10^{-5} (lattice), 10^{-6} (fiber)
Primary algorithm	Newton
Recovery on failure	sub-steps 5/20/50 with Newton → KrylovNewton → NewtonLineSearch
Base shear	$-\sum$ horizontal base reactions

5.2 Dynamic (seismic time history)

After the held-gravity stage, the base is driven by a **UniformExcitation** (ground acceleration record) along x . Two records are used: a harmonic acceleration tuned to resonance with the structure’s fundamental period T_1 , and the recorded El Centro 1940 NS ground motion. The record intensity is tuned on the (inexpensive) fiber column to a target peak roof drift; the *same* scaled record then drives the lattice.

In the *linear* cases all materials are **Elastic**, so both models stay linear at any amplitude. In the *nonlinear* cases the intensity is set so the peak roof drift (about 1.2 in) exceeds the effective yield drift of both models — the elastic limit is ≈ 0.5 in for the fiber column and ≈ 0.6 in for the lattice (peak shear $\div K_0$; see the pushover curves in Section 6.1) — so both the lattice and the fiber beam–column respond in the nonlinear range, leaving a permanent residual drift (Table 16).

Table 12: Dynamic analysis parameters (both models).

Item	Setting
Constraint handler / numberer / system	Transformation / RCM / BandGeneral
Mass	lumped self-mass + top mass P/g
Damping	Rayleigh, 5% at modes 1 and 2 (mass + committed-stiffness)
Time integrator	HHT, $\alpha = 0.7$ (numerical damping)
Integration step	$\Delta t = 0.01$ s
Convergence test	NormDispIncr , tol 10^{-6} , 50 iterations
Primary algorithm	Newton
Recovery on failure	sub-steps 10/50/200 with Newton → KrylovNewton → NewtonLineSearch → ModifiedNewton
Excitation pattern	UniformExcitation along x

Table 13: Base excitation records.

Record	Points	Δt_{rec} (s)	Peak (g)	Duration (s)
Sine resonance (20 cycles, 100 samples/cycle)	2001	$T_1/100$	1.000	$20 T_1$
El Centro 1940 NS	1559	0.0200	0.319	31.2

6 Results

6.1 Pushover

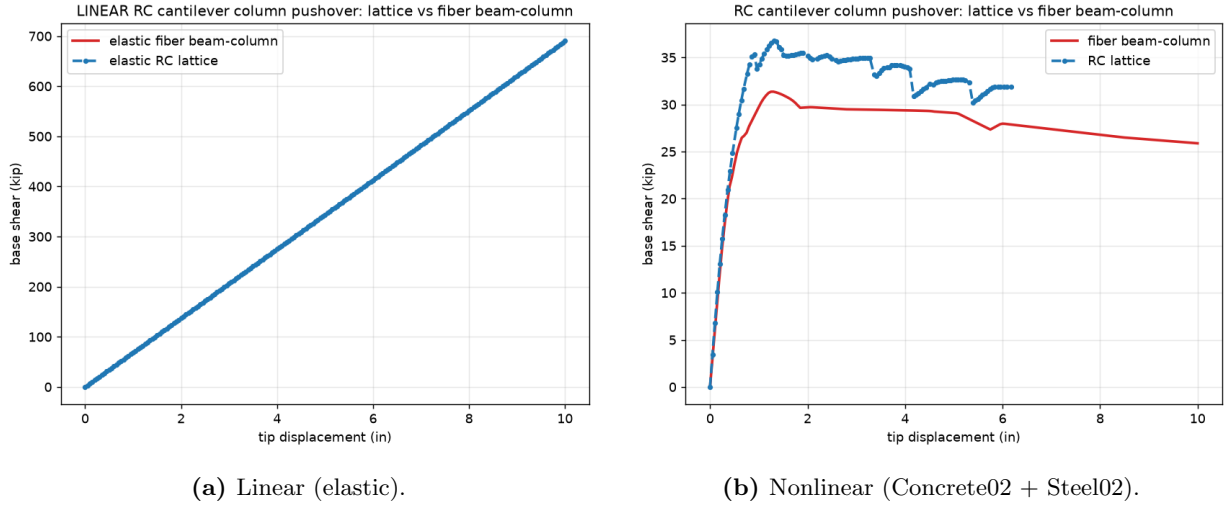


Figure 3: Base shear versus tip displacement: RC lattice (blue, dashed) versus fiber beam-column (red). The linear curves overlap (matched K_0); the nonlinear lattice terminates at 6.17 in where the static solver stops converging.

Table 14: Pushover results.

Configuration	Model	Peak shear (kip)	Max tip disp. (in)	Converged
Linear	Fiber beam-column	686.9 (@10 in)	10.00	yes
	RC lattice	686.9 (@10 in)	10.00	yes
Nonlinear	Fiber beam-column	31.37	10.00	yes
	RC lattice	36.75	6.17	no (terminated)

For each record the response is reported as (i) the roof-drift and base-shear *histories* (both models overlaid) and (ii) the *hysteresis* loops, shown for the RC lattice and the fiber beam-column separately and then overlaid (each loop panel is autoscaled, so the two are legible independently of their force magnitudes).

6.2 Dynamic — linear (elastic)

Sine resonance (scale $0.039\times$).

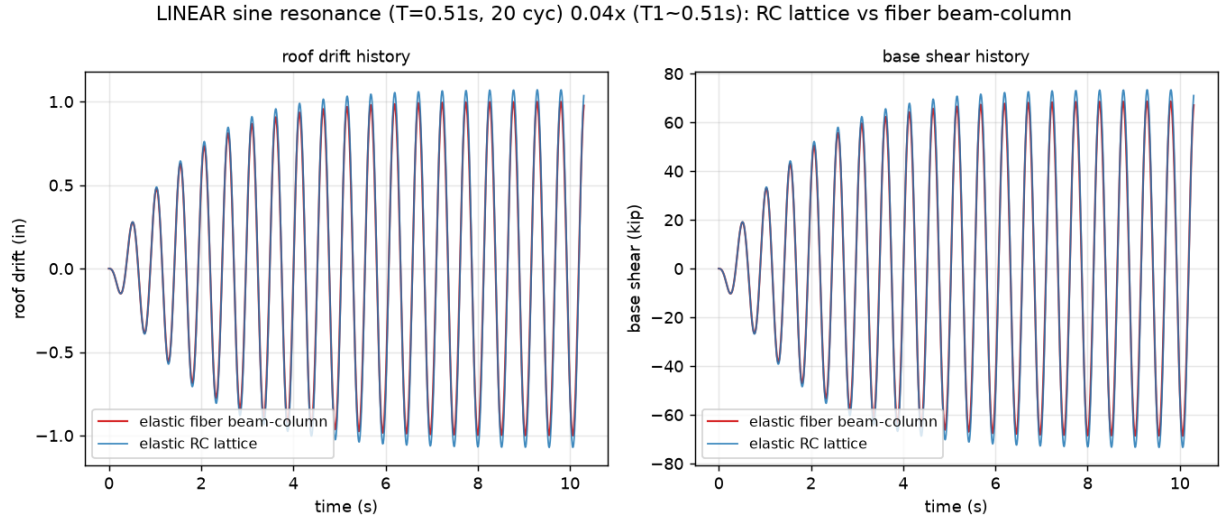


Figure 4: Linear, sine — response histories (roof drift, base shear): RC lattice vs. elastic fiber beam-column.

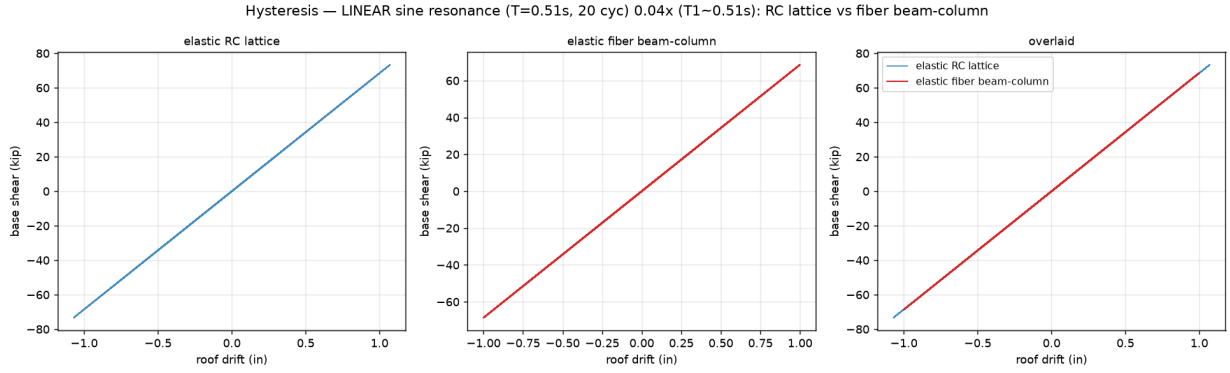


Figure 5: Linear, sine — hysteresis: RC lattice (left), fiber beam-column (middle), and the two overlaid (right).

El Centro 1940 NS (scale $0.412x$).

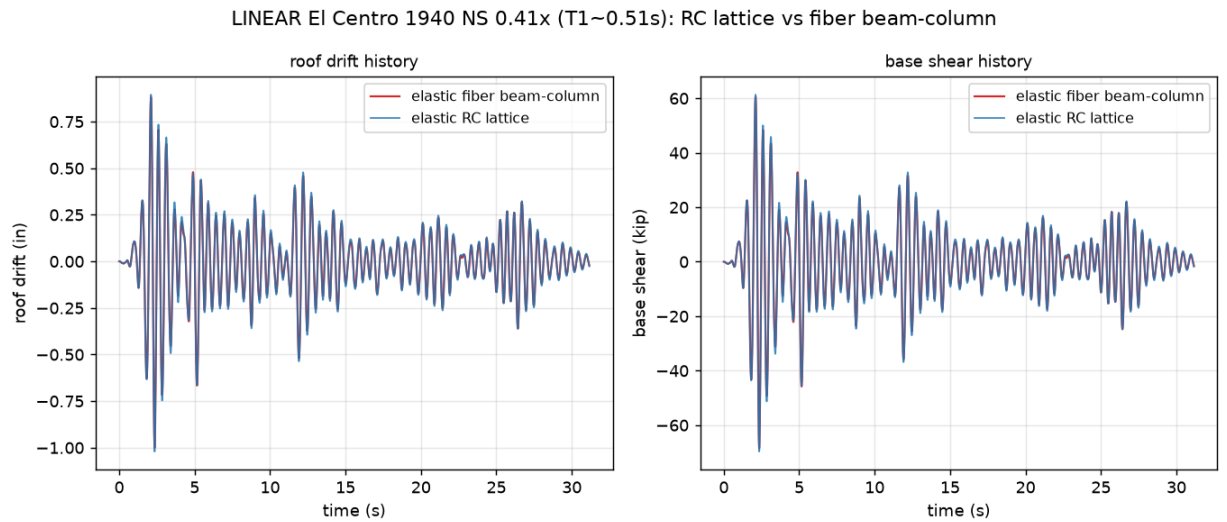


Figure 6: Linear, El Centro — response histories (roof drift, base shear).

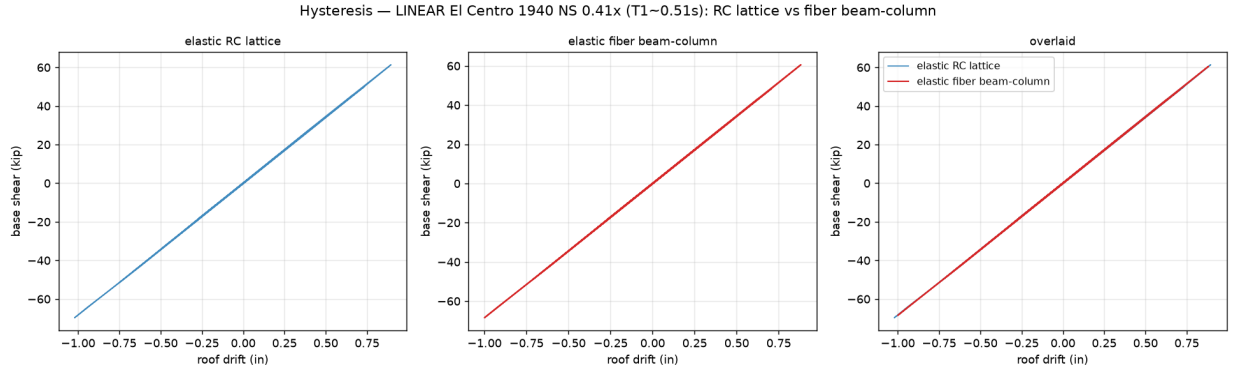


Figure 7: Linear, El Centro — hysteresis: RC lattice (left), fiber beam-column (middle), and the two overlaid (right).

6.3 Dynamic — nonlinear (Concrete02 + Steel02)

Sine resonance (scale $0.145\times$).

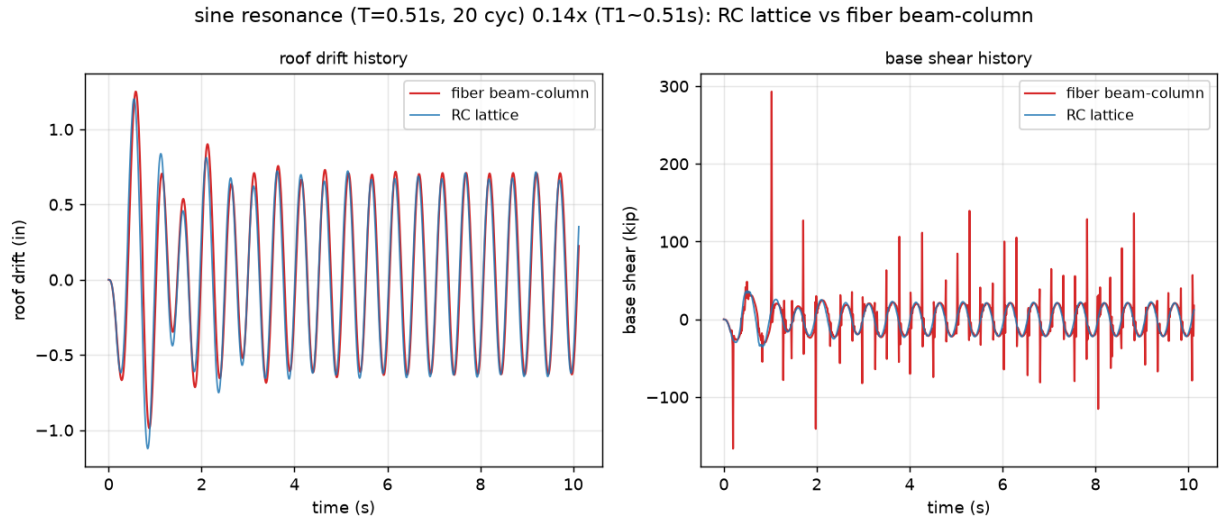


Figure 8: Nonlinear, sine — response histories (roof drift, base shear): RC lattice vs. fiber beam-column.

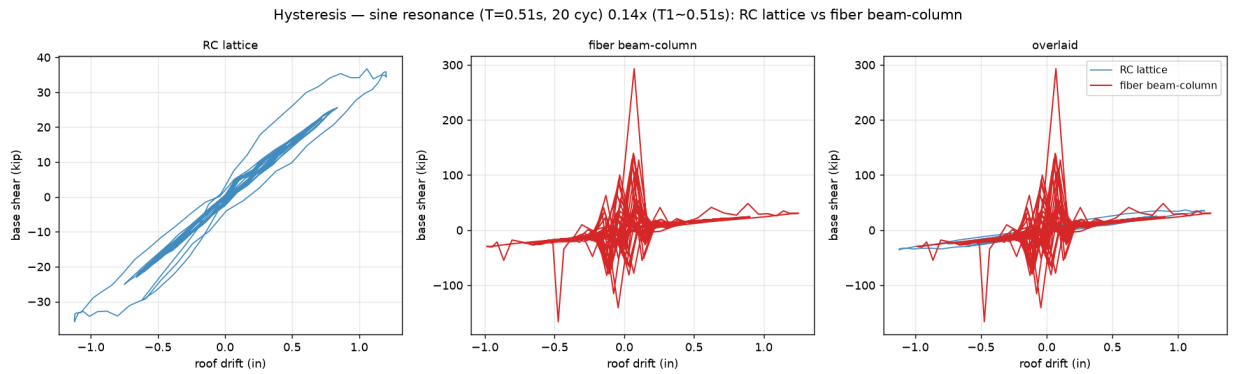


Figure 9: Nonlinear, sine — hysteresis: RC lattice (left, peak ≈ 37 kip), fiber beam-column (middle), and the two overlaid (right). Note the separate vertical scales.

El Centro 1940 NS (scale $0.504\times$).

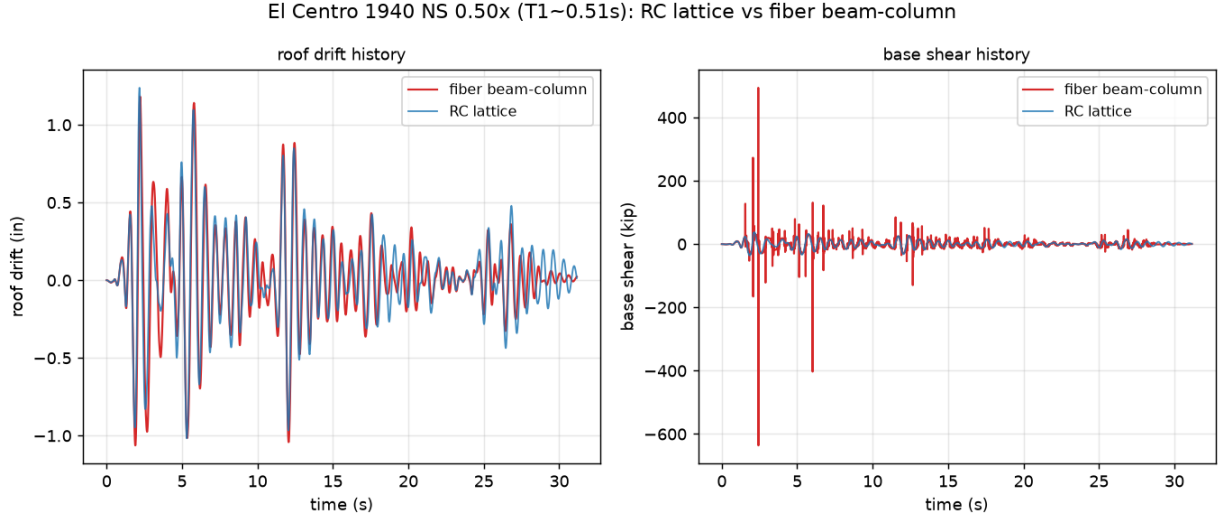


Figure 10: Nonlinear, El Centro — response histories (roof drift, base shear).

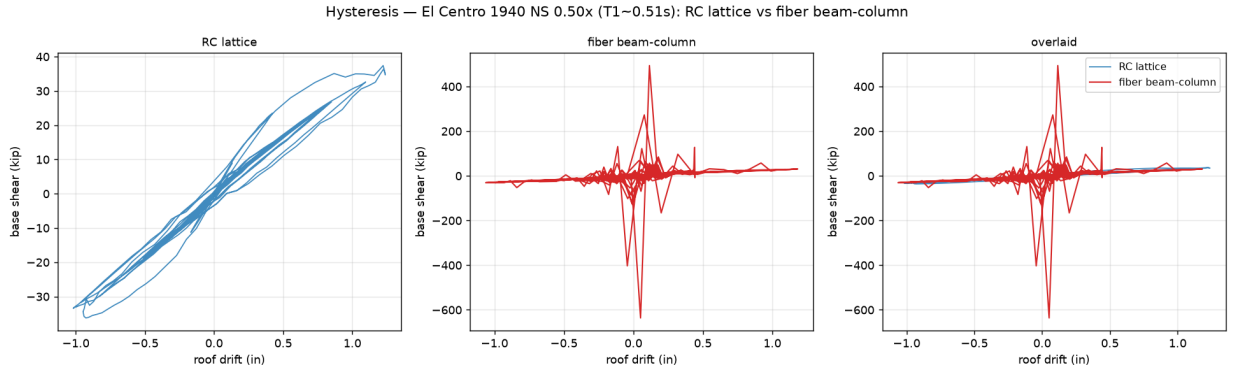


Figure 11: Nonlinear, El Centro — hysteresis: RC lattice (left), fiber beam-column (middle), and the two overlaid (right). Note the separate vertical scales.

Table 15: Dynamic results. Drift is the relative roof displacement; peak $|V|$ is the maximum absolute recorded base shear (includes transient reaction peaks).

Config. / record	Scale	T_1 (s)		Peak drift (in)		Peak $ V $ (kip)	
		Fiber	Lattice	Fiber	Lattice	Fiber	Lattice
Linear, sine	$0.039\times$	0.520	0.515	1.000	1.071	68.7	73.4
Linear, El Centro	$0.412\times$	0.520	0.515	1.000	1.021	68.5	69.7
Nonlinear, sine	$0.145\times$	0.558	0.506	1.251	1.202	293.0	36.7
Nonlinear, El Centro	$0.504\times$	0.558	0.506	1.182	1.238	636.1	37.4

Table 16: Residual (end-of-record) roof drift.

Configuration	Record	Fiber (in)	Lattice (in)
Linear	Sine	+0.976	+1.036
Linear	El Centro	−0.021	−0.027
Nonlinear	Sine	+0.224	+0.353
Nonlinear	El Centro	+0.022	+0.013

A Software and reproduction

Table 17: Software environment (native arm64, macOS).

Component	Version
Python	3.12.13 (arm64)
openseespy	3.8.0.0 (openseespymac 3.8.0.0)
gmsb	4.15.2
numpy / scipy	2.4.6 / 1.17.1
matplotlib	3.11.0

Run from the `src/` directory with the project virtual environment active:

```
python examples/column/pushover.py           # nonlinear pushover
python examples/column/pushover_linear.py     # linear pushover
python examples/column/dynamic.py             --excitation sine
python examples/column/dynamic.py             --excitation elcentro
python examples/column/dynamic_linear.py      --excitation sine
python examples/column/dynamic_linear.py      --excitation elcentro
python examples/doc/make_model_figures.py     # lattice figure
```